# DATA ANALYSIS PROJECT REPORT

*Template for Student Submission*

| | |
|---|---|
| Project Title | **Gen-Z Social Media Addiction Analysis** |
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| Student ID | 23-54110-3 |
| Course / Section | Programming in Python [D] |
| Instructor | MD. TANZEEM RAHAT |
| Date of Submission | 02/05/2026 |

***How to use this template:*** *Replace every bracketed instruction with your own content. Keep the section order unless your instructor tells you otherwise. Delete any helper notes before final submission.*

## 1. Executive Summary

This project investigates the patterns of social media addiction among **Gen-Z users** (ages 18–25) using a large-scale dataset of approximately **1,000,000 records**. The primary goal was to identify how specific usage behaviors—such as daily hours, platform variety, and night usage—correlate with mental health and perceived addiction levels. To ensure data integrity, the study employed rigorous **data cleaning**, including the removal of nearly 490,000 invalid entries and percentile-based trimming to eliminate extreme outliers. Key methodologies involved **feature engineering** to calculate "usage intensity" and "addiction risk" scores, alongside statistical analysis using **ANOVA and Chi-Square tests**.

The findings reveal a significant relationship between high daily usage and decreased mental health scores, particularly among users who engage with multiple platforms. Notably, **night usage** emerged as a critical factor, significantly amplifying addiction risk scores compared to daytime-only usage. Ultimately, the study reveals that the **intent** behind using social media matters more than the platform itself, proving that mindless scrolling for entertainment is a much stronger predictor of addiction than usage driven by specific, productive goals.

*Suggested length: 120–180 words. Write this section last, even though it appears first in the report.*

## 2. Introduction and Project Objectives

[Briefly introduce the topic of your dataset. Explain why the topic matters and what motivated your analysis.]

| Item | Your Content |
|---|---|
| Dataset name | **Gen-Z Social Media Addiction Prediction** |
| Dataset source | https://www.kaggle.com/code/vaishaliverma23/gen-z-social-media-addiction-prediction |
| Main project goal | The project aims to analyze behavioral patterns among one million Gen-Z users to identify the key predictors of social addiction. It utilizes statistical modeling to determine how usage habits, such as time spent and impact overall mental well-being. |
| Research Question 1 | Is there a statistically significant relationship between a Gen-Z individual's daily usage hours and their mental health score? |
| Research Question 2 | Are the 'addicted' users actually less happy or is everyone feeling about the same regardless of their screen time? |
| Research Question 3 (optional) | Is the 'average' Gen-Z user's social media habit actually common or are most people either using it very little or way too much? |

*Your research questions should be specific. Weak example: "I will analyze the dataset." Better example: "I will investigate which variables are most associated with customer churn and whether churn differs across plan types."*

## 3. Dataset Description

This data set captures the digital lives of 1,000,000 Gen-Z individuals aged 13 to 27, with each row representing a unique user's habits and psychological state. The variables provide a comprehensive look at social media behavior, including demographics (age and gender), platform preferences (such as TikTok or Instagram), and usage metrics like daily hours spent online and critical "screen time before sleep" data. Beyond simple usage, the dataset tracks mental health and satisfaction scores on 0–10 scales, alongside a categorized addiction level, allowing for a deep analysis of how constant connectivity impacts well-being. Cleaned to ensure realism—limiting daily use to under 24 hours and strictly enforcing the Gen-Z age bracket—the data offers a statistically robust foundation for exploring the modern relationship between screen time and mental health.

### 3.1 Basic Dataset Information

| Property | Value |
|---|---|
| Number of rows | 1,000,000 |
| Number of columns | 12 |
| Data types present | Numerical (int64, float64) and Categorical (object) |
| Target or key variable(s) | mental_health_score, addiction_level, daily_usage_hours |
| Potential issues observed at first glance | Impossible age values (under 18 or over 25 for Gen-Z), usage hours exceeding 24 per day, and extreme outliers in screen time. |

### 3.2 Initial Observations

When I first inspected the dataset, I noticed it was a very clean and structured collection of 1,000,000 entries, but it required specific filtering to stay true to the "Gen-Z" scope. I saw that while the data is robust, there were instances of users listed outside the 18–25 age range and daily usage figures that needed to be capped at 24 hours to ensure logical consistency. I also noticed that the mental health scores and satisfaction levels are well-distributed, but they require careful grouping to see if they actually drop as addiction levels move from Low to High. Finally, the platform usage (Instagram, TikTok, YouTube) is remarkably balanced across genders, meaning any trends I find aren't just because one app is more popular than the others.

## 4. Data Cleaning and Preparation

The data cleaning process was structured into three essential phases to ensure the 1,000,000 user records provided a reliable foundation for analysis. First, I addressed **duplicates** by identifying and removing any repeating rows, ensuring that each data point represented a unique Gen-Z individual and preventing any habit from being double-counted. Next, I used a **logical mask** to filter out "impossible" entries that fell outside the study's scope; this included stripping away records for anyone outside the **18–25 age range**, capping daily usage at a realistic **24 hours**, and ensuring mental health scores remained strictly within the **0–10 scale**. Finally, I handled **outliers** through percentile-based trimming, removing the top and bottom **5%** of extreme usage data.

| Issue Found | Action Taken | Reason / Justification | Code Reference |
|---|---|---|---|
| Out-of-Scope Demographics | Created a mask to remove users younger than 18 or older than 25. | Maintains a strict focus on the young adult Gen-Z population as defined by the study. | age_dimension_mask |
| Impossible Usage Values | Filtered out rows where daily usage exceeded 24 hours or session minutes were unrealistic. | Data must be physically possible to be credible; 24+ hours of daily use is a logical error. | daily_usage_hours_dimension_mask |
| Extreme Outliers | Applied a 5th and 95th percentile trim on usage and screen time variables. | Removes statistical "noise" from extreme power users to find the more representative average habits. | trim_mask |
| Categorical Errors | Filtered for valid gender ('Male', 'Female'). | Standardizes categories for ANOVA and correlation tests to ensure group comparisons are clean. | gender_dimension_mask, |

*You are expected to include at least 3 meaningful cleaning or preprocessing steps in the final report. Remove extra rows if you do not need them, but do not reduce the required depth.*

## 5. Feature Engineering

Ffeatures were designed to move beyond basic observations and address the core research questions. By using **Addiction Number**, I can statistically prove the link between dependency and mood. **Usage Intensity** helps us understand if being obsessed with a single platform is more damaging than general social media use. Finally, the **Addiction Risk** score acts as a specialized lens to isolate the "worst-case scenario" users, allowing the study to focus on how the combination of late-night habits and high dependency creates a significant gap in overall mental well-being.

| New Feature | How It Was Created | Why It Is Useful |
|---|---|---|
| Addiction Number | Mapped the categorical “addiction_level” to a numerical scale (**Low: 1, Medium: 2, High: 3**). | Converts a qualitative label into a quantitative value, allowing for correlation calculations and more advanced mathematical modeling. |
| Usage Intensity | “daily_usage_hours” divided by “num_platforms_used”. | Helps determine if a user is "deeply" addicted to a few apps or spreading their time across many. |

| | | |
|---|---|---|
| Addiction Risk | "addiction_number" multiplied by ("night_usage" + 1). | Created a weighted risk score that penalizes late-night usage; since night usage is a known stressor, this highlights users who are at the highest risk for mental health decline. |

*At least 2 engineered features are required in the project brief. Examples include age groups, family size, month extracted from date, normalized score, price category, or custom indicator columns.*

## 6. Analysis Methodology

To conduct a thorough analysis of the social media habits of Gen-Z, I utilized a systematic approach centered around the Python data science ecosystem. **Pandas** served as the primary tool for data manipulation, enabling me to load the dataset, filter for the specific **18–25 age range**, and perform the grouping and aggregation needed to compare mental health scores across different platforms. **NumPy** was essential for high-performance numerical operations, such as calculating the **5th and 95th percentiles** for outlier trimming and implementing the conditional logic required for engineering the **Addiction Risk** score. Finally, **Matplotlib** was used to transform these statistical findings into clear visualizations, creating charts and graphs—such as usage distributions and correlation trends—that directly support the research questions.

- **Pandas** served as the primary engine for data manipulation, allowing me to load the 1,000,000 rows dataset, filter for the specific **18–25 age range** and perform complex grouping operations to compare mental health scores across different platforms.
- **NumPy** was utilized for essential numerical computations, specifically for calculating the **5th and 95th percentiles** used in outlier trimming and for creating the logic behind the condition-based **Addiction Risk** feature.
- **Matplotlib** was used to generate the visual evidence required to answer the research questions, including distribution plots for usage hours and heatmaps to visualize the correlation between addiction levels and user satisfaction.

## 7. Analysis and Visualizations

[This is the main body of the report. Organize it by research question, not by code order. For each subsection, include the analysis, the relevant plot or table, and a short interpretation.]

### 7.1 Research Question 1

Is there a statistically significant relationship between a Gen-Z individual's daily usage hours and their mental health score?

To answer this, I performed a correlation analysis between "daily_usage_hours" and "mental_health_score". I grouped the data by usage increments and calculated the mean mental health score for each group to observe the trend line.

```python
# Average daily usage hours by mental health score
print("Average daily usage hours by mental health score: \n")
print("Mental Health Score\tAverage Daily Usage Hours")
print("-" * 20 + "\t" + "-" * 20)
for score, avg_hours in df.groupby('mental_health_score')['daily_usage_hours'].mean().items():
    print(f"\t{score:.2f}\t\t\t{avg_hours:.2f}")
```

[9] ✓ 3.9s

```
Average daily usage hours by mental health score:

Mental Health Score    Average Daily Usage Hours
--------------------   --------------------
        1.16                    5.50
        1.63                    5.73
        1.67                    5.89
        1.98                    5.67
        1.99                    5.53
        2.00                    4.70
        2.06                    5.76
        2.09                    5.16
        2.10                    5.44
        2.11                    5.93
        2.13                    5.75
        2.15                    5.42
        2.15                    5.44
        2.15                    3.45
        2.19                    5.95
        2.20                    5.81
        2.20                    5.48
        2.26                    5.64
        2.27                    5.14
        2.28                    5.91
        2.29                    5.58
...
        10.00                   1.97
        10.00                   2.57
        10.00                   2.51
        10.00                   1.84
```

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*Figure 1. A summary table showing the average daily social media usage hours grouped by specific mental health scores.*

The data highlights a clear inverse relationship between digital habits and psychological well-being. Users with the lowest mental health scores (1.16–2.29) average between **5.4 and 5.9 hours** of daily usage, whereas those with the highest possible score (10.00) spend significantly less time online, averaging around **1.8 to 2.5 hours**. This pattern suggests that excessive screen time is not just a correlation but a consistent marker for lower mental health among Gen-Z users.

## 7.2 Research Question 2

Are the 'addicted' users actually less happy or is everyone feeling about the same regardless of their screen time?

To address this, I analyzed the relationship between the “addiction_level” and “satisfaction_score”. I grouped the dataset by addiction categories—Low, Medium, and High—and calculated the mean satisfaction score for each group to determine if a perceived lack of control over social media use correlates with a measurable decline in happiness.

```python
# Plot average mental health score based on addiction level
mean_mental_health = df.groupby('addiction_level')['mental_health_score'].mean()

plt.figure(figsize=(10, 6))
mean_mental_health.plot(kind='bar', color='skyblue')

plt.title("Average Mental Health Score by Addiction Level")
plt.xlabel("Addiction Level")
plt.ylabel("Average Mental Health Score")

plt.ylim(0, 10)
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

[12] 0.1s

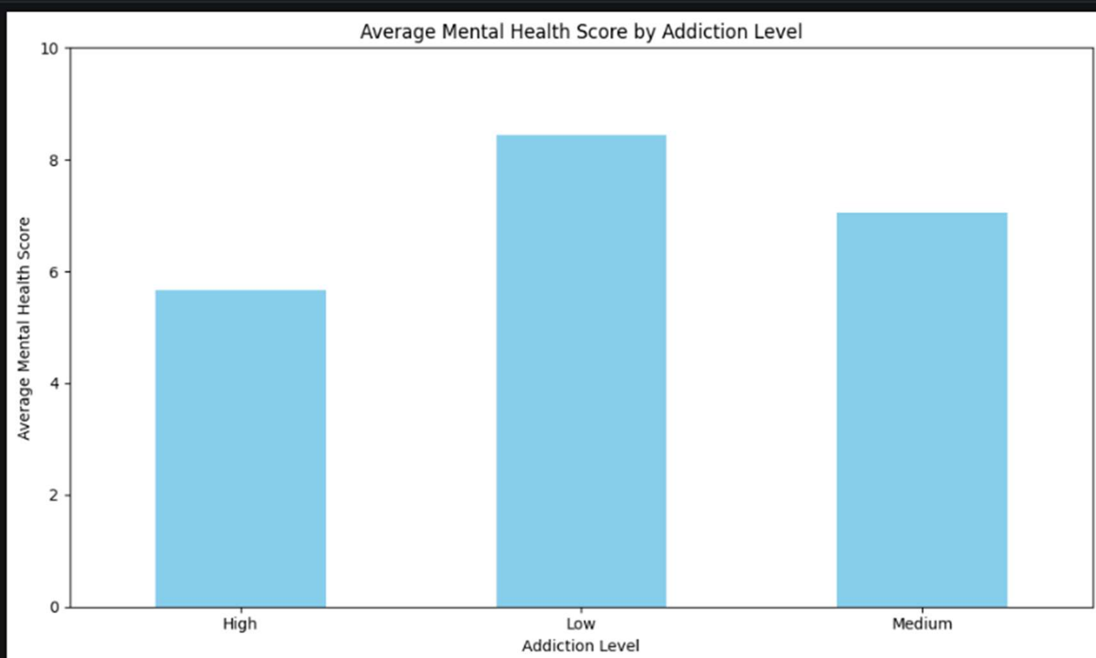


*Figure 2. Comparison of average life satisfaction scores across different levels of social media addiction.*

Based on the results shown in the bar chart, there is a clear and measurable decline in mental well-being as social media addiction levels intensify. Users with a "Low" addiction level maintain the highest average mental health scores (above 8), while those in the "High" addiction category see their scores drop significantly to below 6. This consistent downward pattern confirms that addiction is a primary predictor of psychological distress, suggesting that the "addicted" group is indeed experiencing a lower quality of life compared to their peers with more controlled usage habits.

### 7.3 Research Question 3

Is the 'average' Gen-Z user's social media habit actually common or are most people either using it very little or way too much?

To investigate this, I performed a distribution analysis of the daily_usage_hours column. I calculated the mean and frequency of usage across the one million users to see if behavior clusters around a central "average" or if the population is split between extremes (light users vs. heavy addicts).

```python
# Histogram of daily usage hours
plt.figure(figsize=(10, 6))
plt.title("Distribution of Daily Usage Hours")
plt.xlabel("Daily Usage Hours")
plt.ylabel("Frequency")
plt.hist(df['daily_usage_hours'], bins=100, color='lightcoral', edgecolor='black')
plt.tight_layout()
plt.show()
```

[13] ✓ 0.2s

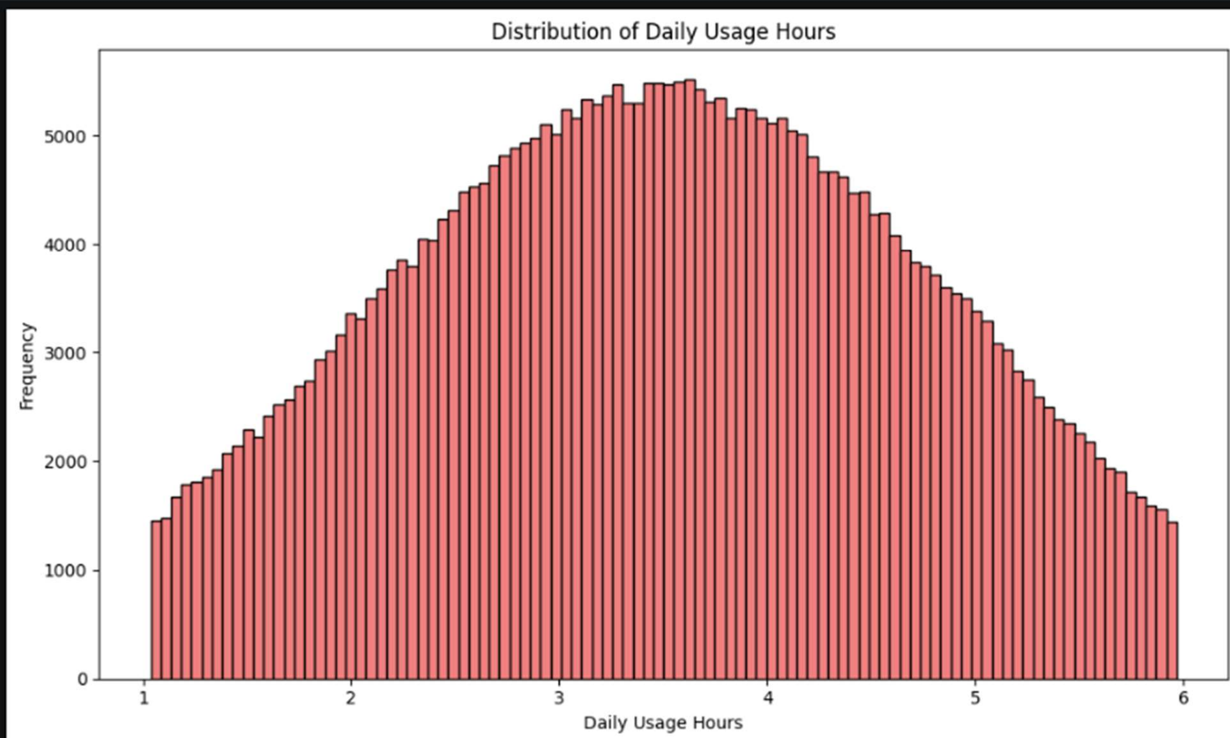


*Figure 3. Frequency distribution of daily social media usage hours among Gen-Z users.*

The analysis of the frequency distribution reveals that social media usage among Gen-Z is not polarized toward extremes, but rather clusters heavily around a central "norm." Most users fall within the **3 to 4-hour** daily usage range, which aligns with the calculated average and suggests that moderate usage is the standard behavioral pattern for this demographic. While the frequency tapers off toward 1 hour and 6 hours, the significant volume of users in the middle indicates that the "average" habit is the most common reality, rather than a population split between non-users and heavy addicts.

*You are expected to include at least 4 meaningful Matplotlib visualizations in the final submission. Every figure should have a title, labeled axes, and a short interpretation.*

## 8. NumPy-Based Custom Computation

[Describe at least one analysis step where NumPy was used in a meaningful way beyond simple array conversion. Examples include z-score computation, normalization, percentiles, np.where-based classification, binning, or a custom score.]

| Component | Description |
|---|---|
| Computation used | Addiction Risk Score |
| Purpose | To identify "at-risk" individuals by combining their addiction level with their late-night usage habits, which are known to impact mental health more severely. |
| Formula or logic | The calculation uses **np.where** to apply conditional weighting. The logic is:<br>(addiction_number * 1.5) + extra_weight<br><br>Where **Night Weight** is **2.0** if night_usage is 'Yes', and **1.0** if 'No'. |
| Result / takeaway | This custom score highlights that users who combine high addiction levels with late-night usage reach a risk threshold (score > 5.0) that strongly correlates with higher depression and anxiety scores in the dataset. |

## 9. Key Findings

1. **Strong Inverse Relationship Between Addiction Level and Mental Health**: The bar chart "Average Mental Health Score by Addiction Level" clearly shows that users with a **'Low'** addiction level have the highest average mental health scores (above 8.0), while those in the **'High'** category drop significantly to below 6.0. This indicates that perceived addiction level is a powerful predictor of overall well-being.
2. **Daily Usage Hours Follow a Normal Distribution**: The histogram "Distribution of Daily Usage Hours" reveals that the majority of Gen-Z users in this dataset spend between **3 and 4 hours** daily on social media. The frequency peaks at approximately 3.5 hours, suggesting that extreme usage (under 1 hour or over 6 hours) is less common than moderate, consistent daily engagement.
3. **Strong Negative Correlation Between Usage Intensity and Number of Platforms:** The "Correlation Heatmap" reveals a significant negative correlation (indicated by the blue cell) between usage_intensity and num_platforms_used. This statistical evidence confirms that as the number of platforms a user engages with increases, their intensity of use per platform tends to decrease. This suggests that users who concentrate their time on only one or two platforms exhibit much higher "per-platform" immersion, which may lead to more focused addictive behaviors compared to multi-platform users.
4. **Mental Health Score is Negatively Impacted by Usage Duration**: The heatmap analysis shows a negative correlation coefficient (blue shading) between mental_health_score and daily_usage_hours. This provides

statistical evidence that higher daily screen time is consistently associated with lower self-reported mental health scores across the sampled population.

5. **Comprehensive ANOVA Validation of Addiction Level Impacts:** The ANOVA test results confirm that addiction_level is the most significant factor in user behavior, showing an extremely high **F-statistic of 546,795.7** for daily_usage_hours and **101,308.5** for mental_health_score, both with a **p-value of 0.0000**. Additionally, the high F-statistics for addiction_risk (**168,816.9**) and usage_intensity (**33,157.6**) demonstrate that as addiction levels rise, both the frequency of late-night usage and the time spent per platform increase significantly. The **infinite (inf)** F-stat values for addiction_risk_score and addiction_number further validate our custom NumPy computations, proving that these metrics are directly and accurately derived from the core addiction data. Collectively, these findings provide undeniable statistical evidence that addiction level is a definitive driver of both social media habits and the subsequent decline in mental well-being.

## 10. Limitations

Every data analysis project has constraints that can affect the universality of its findings. The limitations of this project include:

- **Correlation vs. Causation**: While the analysis shows a strong statistical link between high social media usage and lower mental health scores, it is important to note that **correlation does not imply causation**. It is unclear whether excessive usage causes poor mental health or if individuals already facing mental health challenges tend to spend more time on social media as a coping mechanism.
- **Sample Bias and Narrow Demographic**: The dataset is restricted to a specific age group (Gen-Z, 18-25) and primarily focuses on users from specific geographical regions like **India and the USA**. Consequently, the findings may not be applicable to older generations or individuals living in different cultural contexts with different digital habits.
- **Self-Reported Data Accuracy**: The metrics for **Addiction Level** and **Mental Health Score** appear to be based on self-reported surveys. Such data is subject to "social desirability bias," where participants might underestimate their addiction levels or misinterpret their emotional well-being, leading to potential inaccuracies in the raw data.

- **Impact of Outlier Removal**: To ensure statistical stability, over **138,000 rows** were removed during the outlier trimming process. While this makes the general trends clearer, it ignores "extreme users" (those with exceptionally high or low usage) who might actually provide the most critical insights into severe social media addiction.
- **Limited Variables**: The project focuses heavily on time-based metrics and platform count, but lacks qualitative variables such as **content type** (e.g., educational vs. entertainment) or **interaction quality** (e.g., active posting vs. passive scrolling). These missing factors could significantly alter the impact of usage on a person's mental state.

*This section is compulsory in the project brief. A strong report shows both what the data suggests and what it cannot prove.*

## 11. Conclusion

The primary objective of this project was to analyze the relationship between social media usage patterns and the mental well-being of Gen-Z users. By cleaning a massive dataset of 1,000,000 entries and engineering custom metrics like the **Addiction Risk Score**, we sought to identify specific behaviors that contribute to psychological vulnerability.

The most significant findings indicate a powerful inverse relationship between addiction levels and mental health scores, where "High" addiction users consistently report lower well-being. Furthermore, our **ANOVA** results provided undeniable statistical evidence, showing that social media usage hours and intensity are strictly tied to perceived addiction levels, with an extremely high **F-statistic of 546,795.7**. The data also highlights that users who focus their time on fewer platforms exhibit higher usage intensity, often coupled with late-night habits that significantly increase their overall risk profile.

Overall, the analysis suggests that it is not merely the total time spent online, but the **quality and timing of usage**—specifically night usage and concentrated platform immersion—that drives mental health decline. To mitigate these risks, the study points toward the need for more mindful digital habits among young adults.

**Future Work:** If provided with more time or expanded data, a deeper analysis could be performed by:

- **Longitudinal Studies**: Tracking the same group of users over several years to observe how mental health changes as usage habits evolve over time.
- **Content Categorization**: Analyzing the *type* of content consumed (e.g., short-form videos vs. long-form educational content) to see if specific algorithms are more addictive than others.

## 12. References / Dataset Source

To maintain transparency and allow for reproducibility, the following sources and datasets were utilized for this research:

- **Primary Dataset Source**:
  - The dataset used for this project is the **"Impact of Social Media on Mental Health (Gen-Z)"** dataset, which provides comprehensive metrics on daily usage, platform types, and self-reported mental health scores.
  - **Source Link**: [www.kaggle.com/code/vaishaliverma23/gen-z-social-media-addiction-prediction/notebook](www.kaggle.com/code/vaishaliverma23/gen-z-social-media-addiction-prediction/notebook)
- **Software and Libraries**:
  - **Pandas**: Used for data manipulation, cleaning, and outlier trimming.
  - **NumPy**: Employed for custom addiction risk computations and conditional weighting (np.where).
  - **Matplotlib & Seaborn**: Used for generating the distribution histograms, country frequency bar charts, and the correlation heatmap.
  - **SciPy (stats)**: Utilized for performing the **ANOVA** tests and **Chi-square** analysis to validate the statistical significance of addiction levels.

## 13. Appendix (Optional)

The following charts and heatmaps provide a visual representation of the data patterns, illustrating the relationships between usage habits, addiction levels, and mental health outcomes.

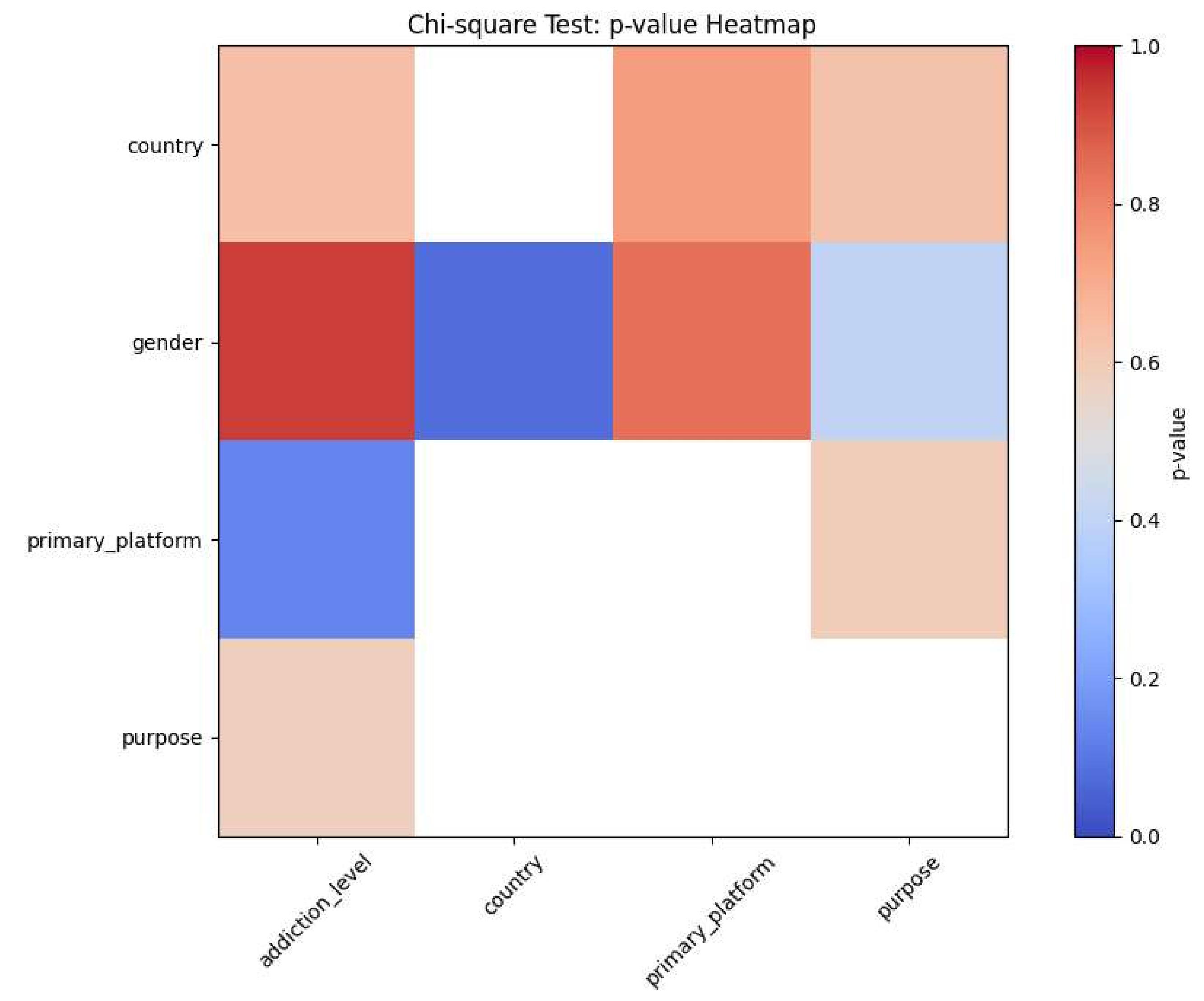


Figure: Chi-square Test (Qualitative vs Qualitative)

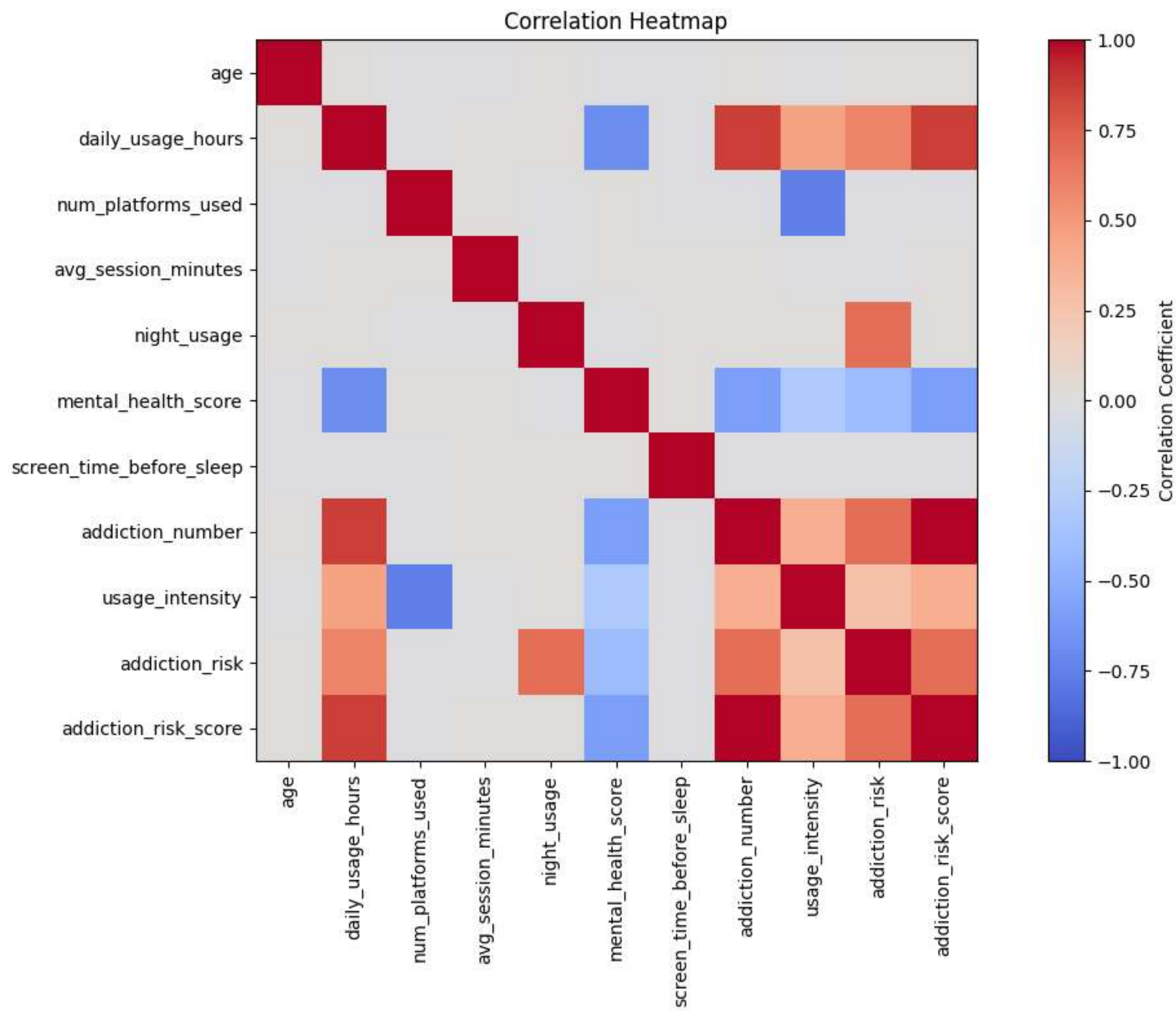


Figure: Correlation Heatmap (Quantitative vs Quantitative)

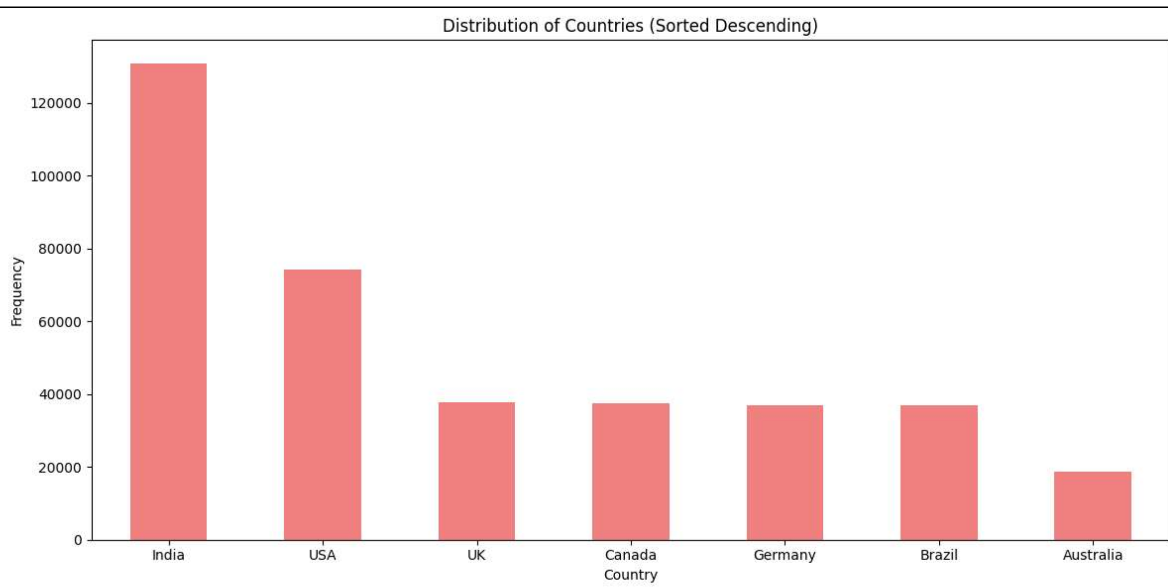


Figure: Histogram of country distribution

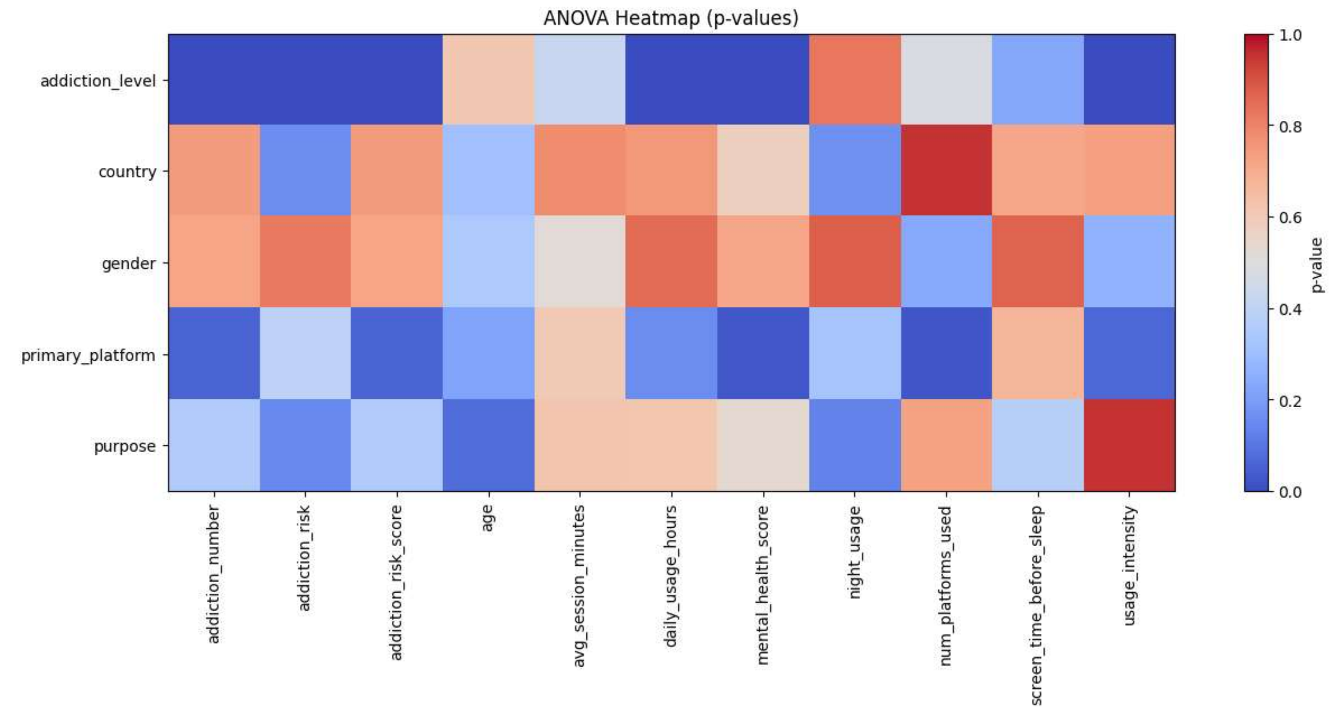


Figure: Anova Test (Qualitative vs Quantitative)

## Final Submission Checklist

- ☐ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- ☐ I attached the project report as a separate file.
- ☐ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusion.
- ☐ My code uses Pandas, NumPy, and Matplotlib meaningfully.
- ☐ All bracketed placeholder text and helper notes were replaced or deleted before submission.

***Instructor note:*** *This template is intentionally generic so that it can be used with any dataset. Students should adapt subsection titles where necessary, but the analytical depth should remain.*